ITCS216 Fundamentals of Database Systems (1/2026)

Workshop: Mapping EERD to the Relational Model

Student ID: _____ Name: _____ Section: _____

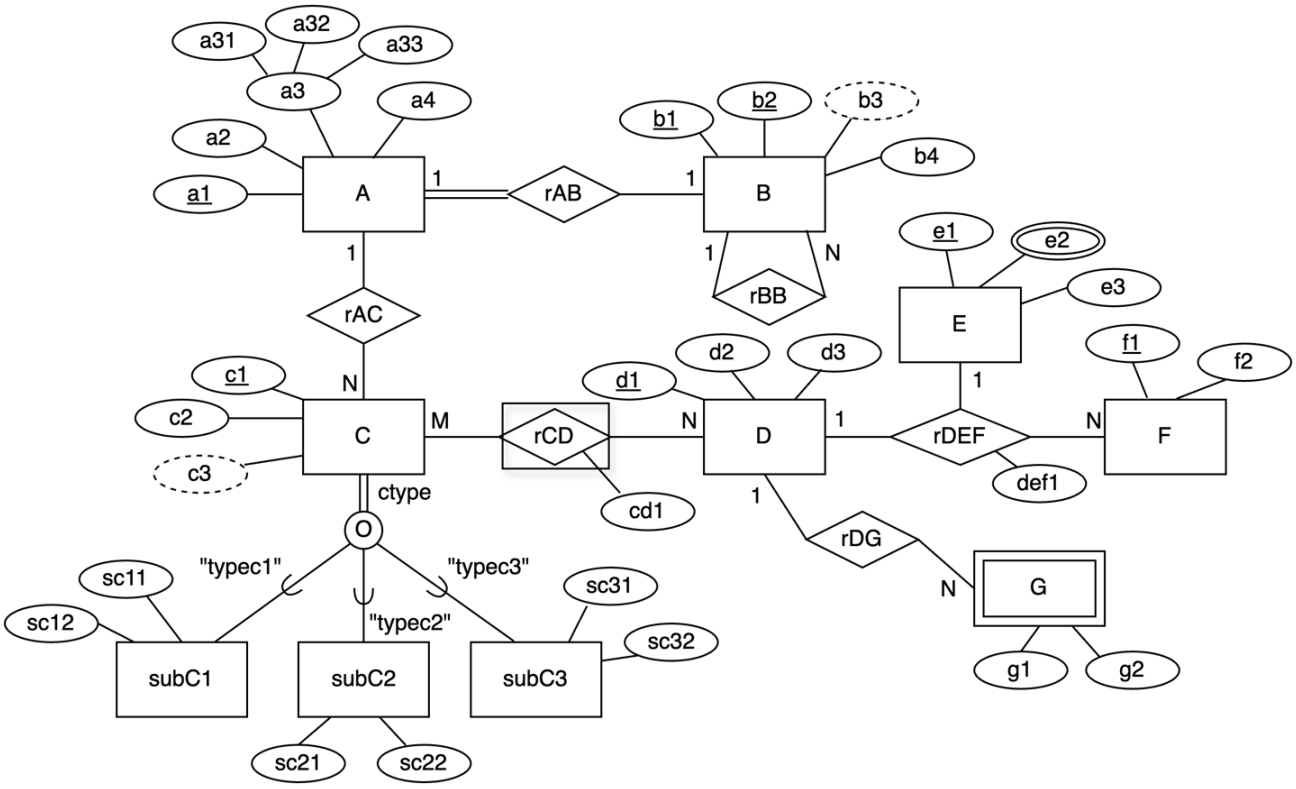
Learning objective

Transform a given EERD into a relational model by applying the 8-step transformation procedure and explaining key mapping decisions.

Instructions

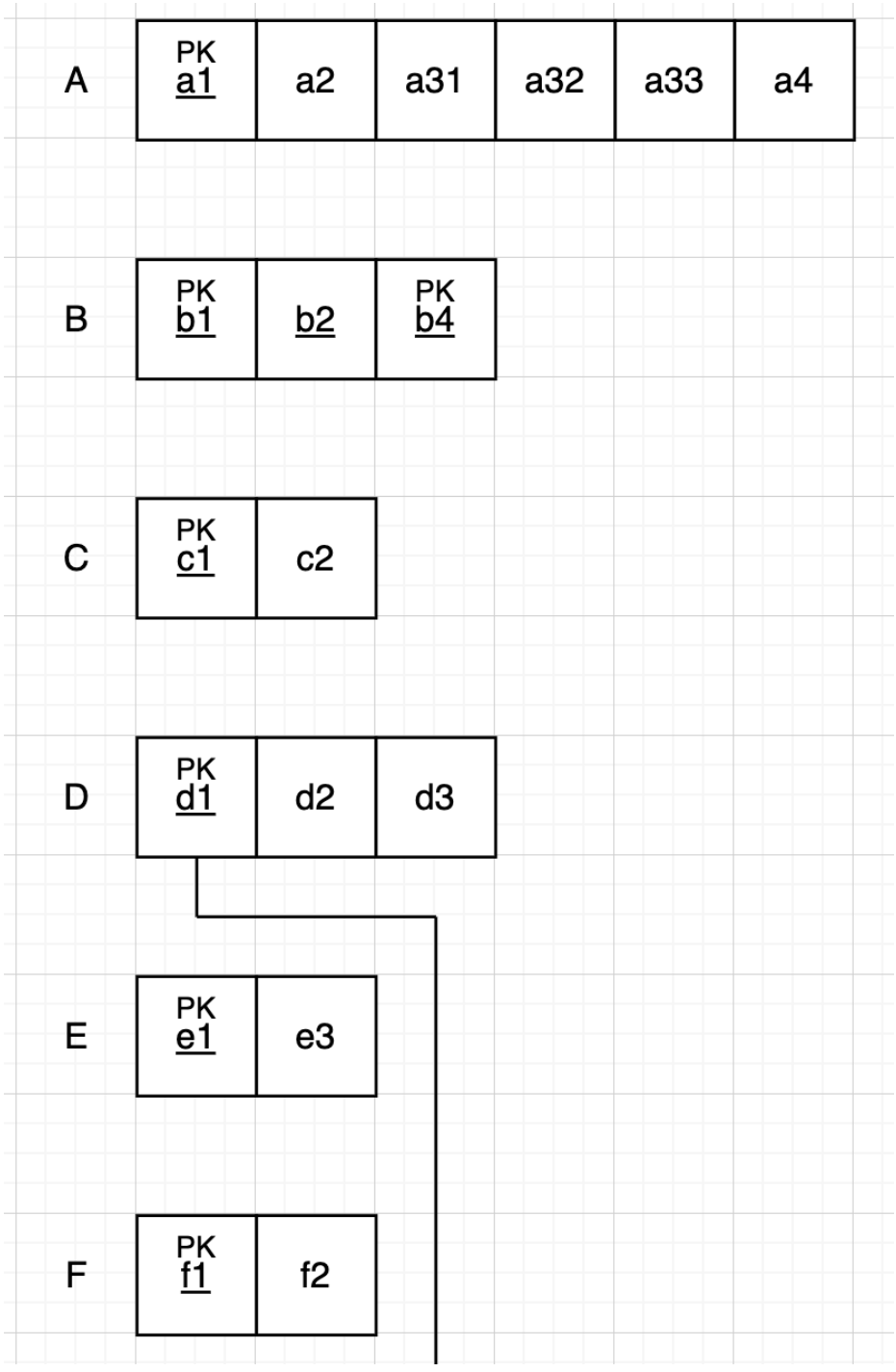
1. Work through Steps 1-8 in order.
2. Show the schema after each applicable step; do not jump directly to the final answer.
3. Clearly mark PK and FK attributes.
4. If a step does not apply, write "Not applicable" and briefly state why.
5. For Step 8, circle one option and justify your choice.
6. Submit by the end of the class.

Given EERD



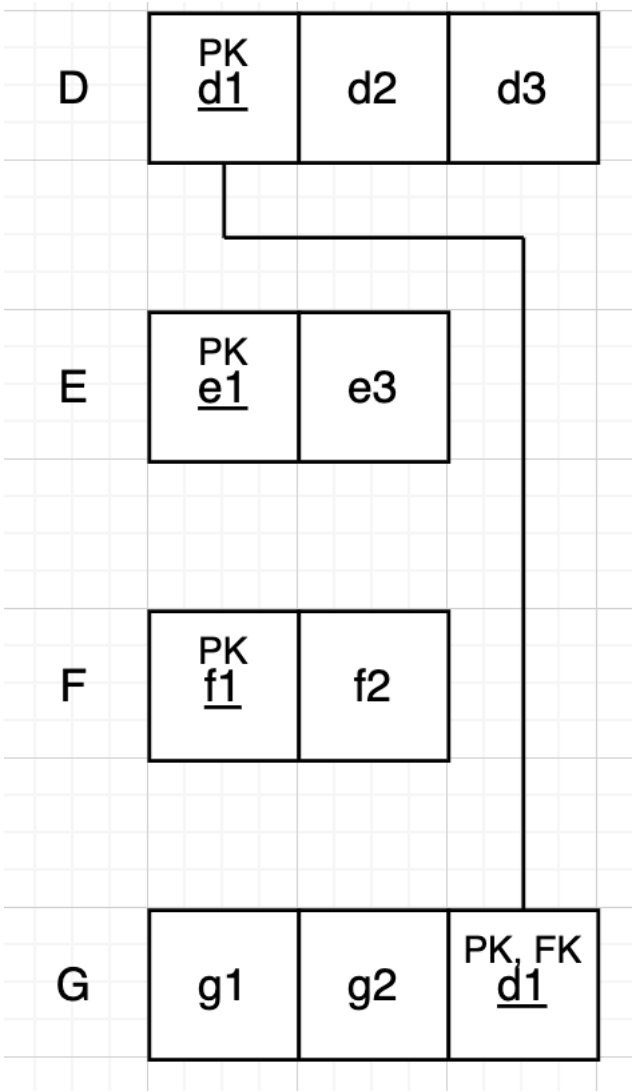
Step 1 - Transform regular entities

- Create one relation for each regular entity.
- Map simple attributes directly; replace composite attributes with their component attributes.
- Identify the primary key of each relation.



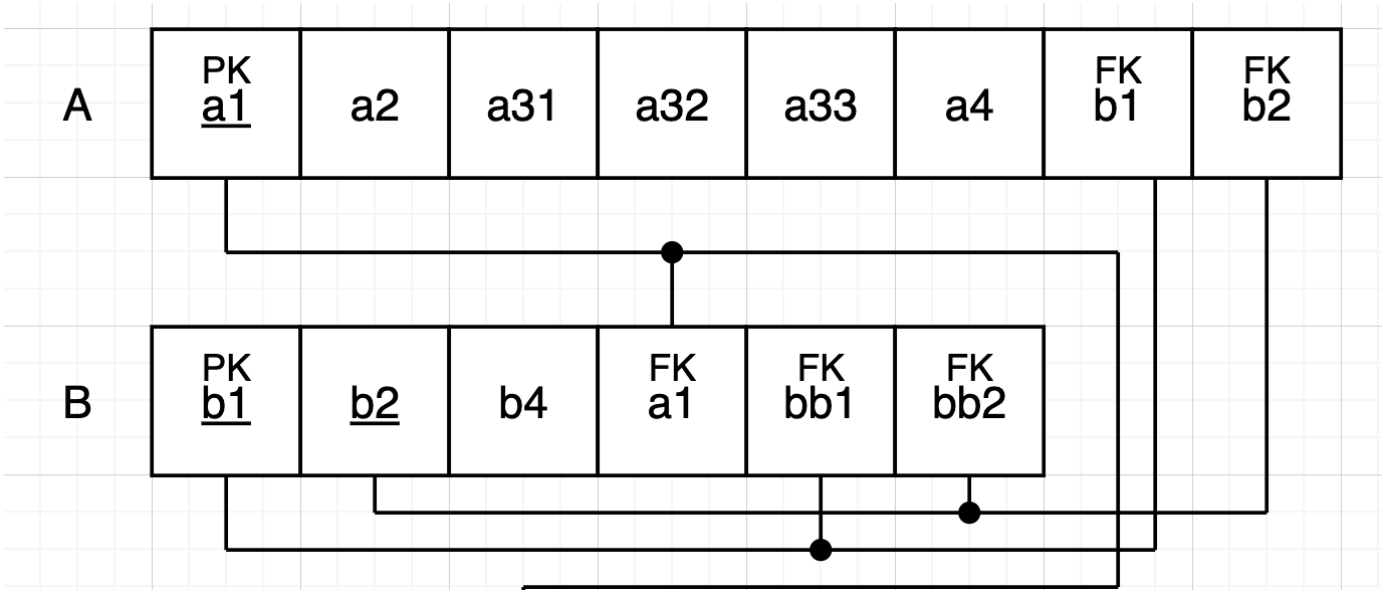
Step 2 - Transform weak entities

- Identify the weak entity and its owner/parent entity.
- Include the parent's PK as an FK in the weak-entity relation.
- State the PK of the weak-entity relation.



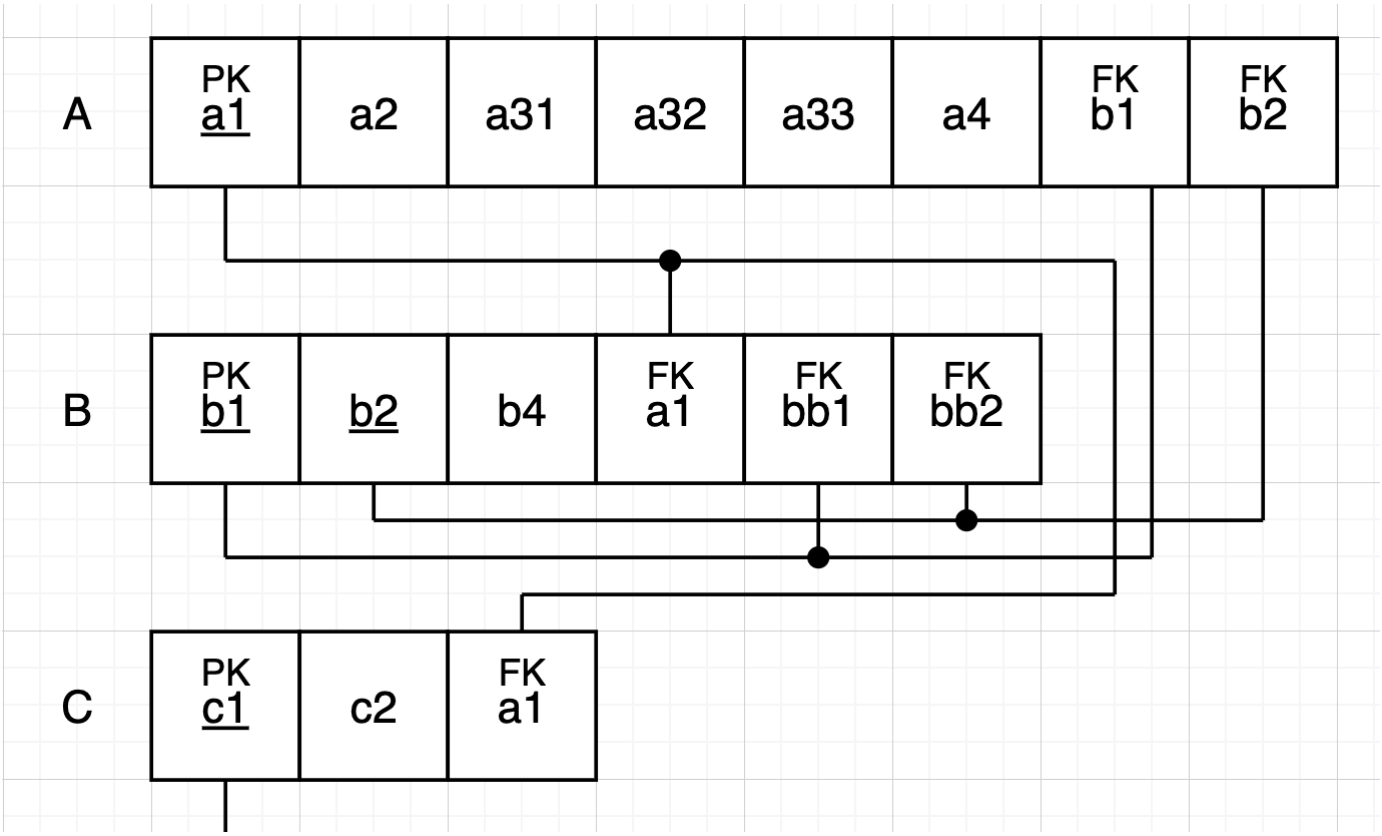
Step 3 - Transform binary 1:1 relationships

- Identify the 1:1 relationship(s).
- Choose which relation receives the foreign key.



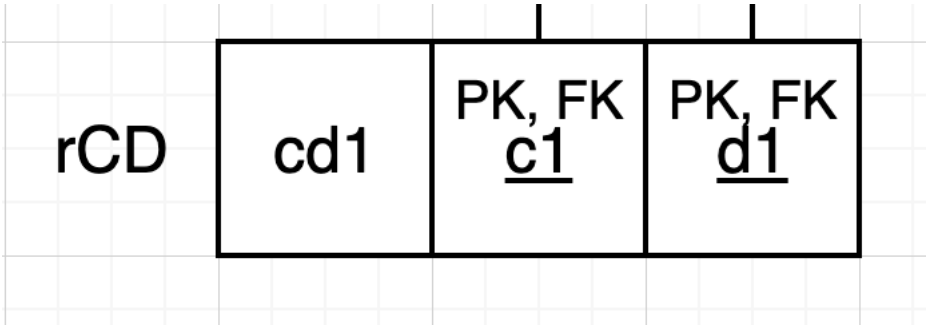
Step 4 - Transform binary 1:N relationships

- Identify each 1:N relationship.
- Place the PK of the "1" side as an FK in the relation on the "N" side.
- Include any relationship attributes, if applicable.



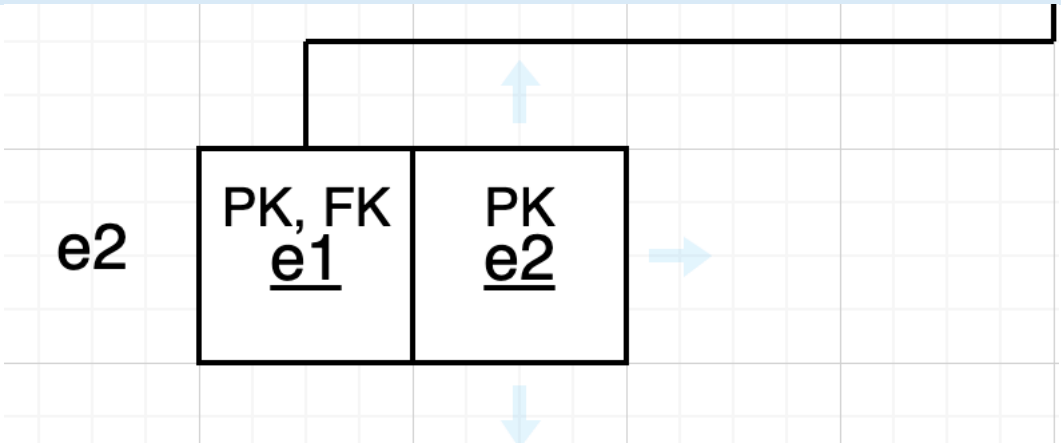
Step 5 - Transform binary M:N relationships

- Create a new relation for each M:N relationship.
- Include the PKs of participating entities as FKs and identify the resulting PK.
- Include relationship attributes, if any.



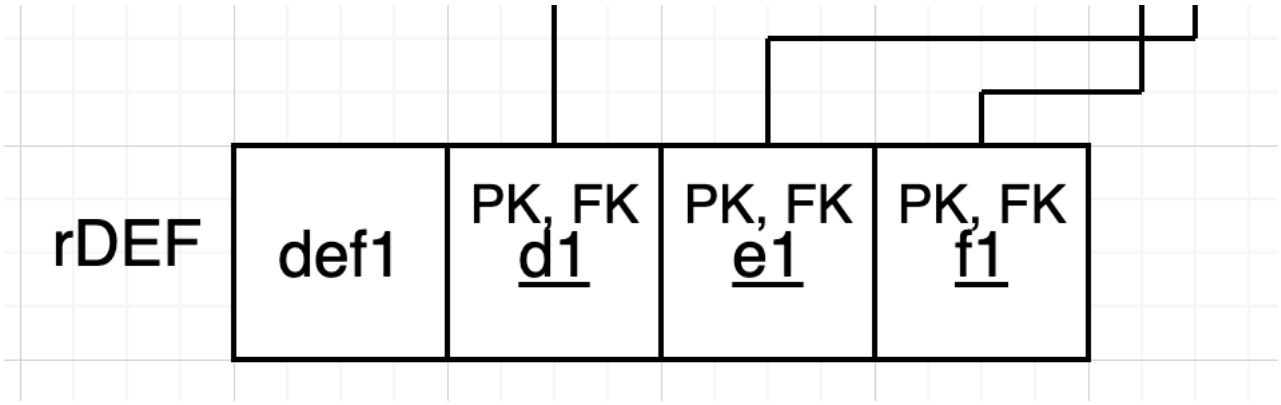
Step 6 - Transform multivalued attributes

- Create a new relation for each multivalued attribute.
- Include the owner entity's PK as an FK and state the PK of the new relation.



Step 7 - Transform N-ary relationships

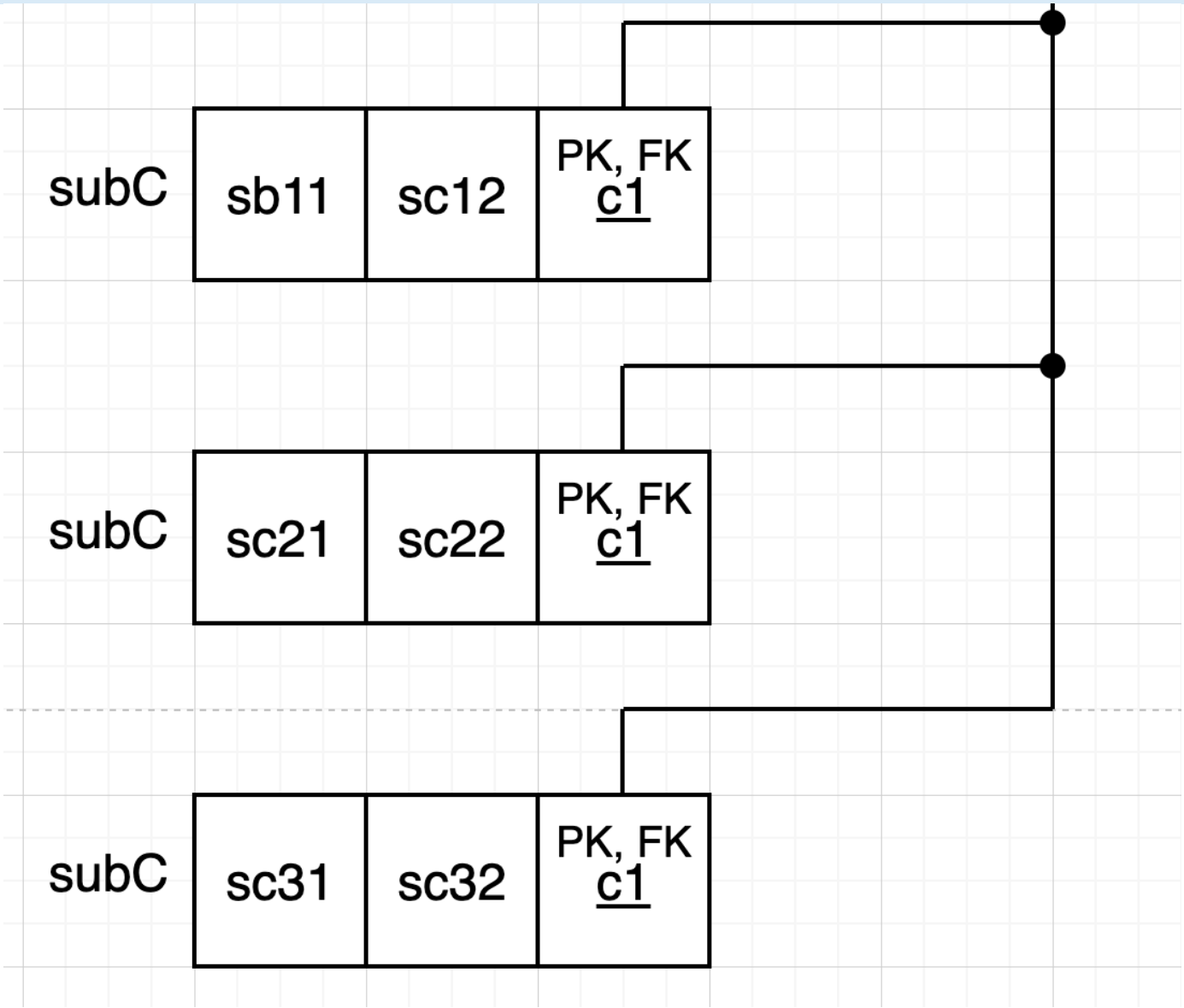
- Create a new relation for the N-ary relationship.
- Include the PKs of all participating entities as FKs.
- Determine the PK based on the cardinality constraints and include any relationship attributes.



Step 8 - Transform specialization / generalization

Circle one mapping option: 8A 8B 8C 8D

- State the specialization constraints you observe.
- Briefly explain why the selected option is appropriate.
- Write the resulting relation(s) and clearly mark PK/FK attributes.



Final relational model

Write the complete final schema. Clearly mark PK and FK attributes in every relation.

